Philippe Decoste

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Roboticist with industry experience building real-time vision systems. Published author at top robotics conference. Background in aerospace and current focus on applied mathematics for state estimation.

EDUCATION

Master of Applied Science in Mechanical Engineering **Sept. 2021 – Dec. 2023**

McGill University, Montreal, Canada — DECAR Systems Group§ , Advisor: James Richard Forbes§

- Research in autonomous navigation and Bayesian state estimation in challenging environments using thermal cameras.
- Author on a paper§ presented at the International Conference on Robotics and Automation (ICRA), the top conference in the field of robotics, for my contributions to a robust homography-based navigation method employing Lie groups.
- Built core expertise in optimization and differential geometry to derive Lie group-based Kalman filters and batch solvers for state-of-the-art SLAM, incorporating covariance quantification for robust navigation in uncertain environments.
- Classes covered subjects such as model predictive control, machine learning, computer vision and optimization proofs; projects include building an optimal path planner in unknown environments and a numerical optimization library.

Bachelor of Engineering, Aerospace, *Space Systems Design, Co-op Option* **Sept. 2016 – April 2021**

Carleton University, Ottawa, Canada

- Graduated with high distinction and obtained senate medal for high academic achievement.
- Design and analysis of aircraft, spacecraft, and rockets. Emphasis on performance & control theory of flights and orbits.
- Classes and projects included orbital mechanics, satellite design and control, communication links, trans-atmospheric propulsion.

WORK EXPERIENCE

Perception Systems Applied Scientist **February 2024 – August 2024 (Contract)**

ARA Robotique, Montreal, Canada

- Engineered a state estimation framework for long-range (1.5+ km) dynamic object tracking, fusing monocular vision with GNSS data to provide uncertainty-informed position and velocity estimates, achieving <5% error in field tests.
- Integrated the algorithm into an aerial drone, using Lie theory to clearly and efficiently model dynamic frames of reference for real-time localization.
- Improved the project's documentation process by providing clear and concise explanations of complex mathematical concepts with a switch to LaTeX-based derivations and diagrams, leading to quicker onboarding of new team members.
- Developed the algorithm and its ROS integration in C++ / Python through collaboration on Git with proper CI/CD practices and unit/integration tested code achieving increased edge case coverage.

Spacecraft GNC Researcher **May 2020 – August 2020**

Carleton University – Spacecraft Robotics and Control Laboratory, Ottawa, Canada

- Worked to develop and enhance the ASTRO multi-agent collision avoidance and trajectory optimization algorithm proposed by NASA astronaut Gregory Chamitoff under Professor Steve Ulrich.
- Investigated state of the art path-planning literature resulting in proposed improvements such as collaborative trajectory optimization protocols to enhance the solving efficiency of multi-UAV pathing and decrease computation time.
- Presented to industry professionals from Carleton University and NASA's Jet Propulsion Laboratory with great feedback.

Co-op Student in Aerostructures (Stress & Design) and Airworthiness **May 2019 – March 2020**

Bombardier Aerospace / DeHavilland Aircraft of Canada Ltd. – Q400 Program, Toronto, Ontario

- Designed and 3D modeled structural components using CATIA, integrating them with existing aircraft structure to handle new loading requirements of aircraft modification project; interpreted relevant technical drawings.
- Led meetings with technical experts to assess impact on aircraft safety, reaching solutions for risk mitigation multilaterally across departments.

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Space Science and Technology Intern **May 2018 – August 2018**

Canadian Space Agency (CSA) – David Florida Laboratory (DFL) Satellite Testing Facility, Nepean, Ontario

- Monitored and documented ongoing and historic spacecraft Assembly, Integration, and Testing (AIT) protocols and campaigns globally to author a 70-page foresight report on future AIT infrastructure requirements for lunar exploration.
- Provided actionable recommendations to leadership on the CSA's technical infrastructure readiness for interplanetary mission qualification (vibration, thermal vacuum, environmental, EMI/EMC).
- Supported environmental testing of CubeSats as part of the Canadian Cubesat Challenge, including thermal vacuum, electromagnetic compatibility, and vibration qualification campaigns.
- Gained foundational exposure to planetary science and deep space mission architectures, reinforcing coursework in spacecraft systems and design.

SKILLS

Software: ROS • OpenCV • OpenGL • Ansys STK • CATIA • Solid Works • Onshape • Linux • Docker • Git • LaTeX
Programming: Python • C++ • MATLAB • Simulink
Mathematical Tools: Probability and Statistics • Optimization • Computer Vision • SLAM • Control Theory • Path Planning • System Modelling and Simulation

PROJECTS

Hardware integration – **Sensor Fusion Rig (Thesis) -** *ROS, Docker, C++, Computer Vision*

- Integrated a thermal camera–IMU rig with ROS for real-world data collection and visual-inertial sensor fusion validation.
- Performed stereo camera-to-camera and camera-to-imu calibration to ensure sensor head accuracy.

Thermal-Inertial Navigation for GNSS-Denied Extreme Environments (Thesis) - *State estimation, computer vision, simulation, optimization, Python*

- Derived and prototyped a thermal signature-based visual-inertial navigation backend through numerical modeling, path-planning, and simulation in Python, enabling rapid, low-cost iteration prior to hardware deployment.
- Validated the thermal navigation system against motion capture ground truth, significantly outperforming conventional feature-based methods by leveraging a novel adaptation of a perception method from the field of control theory.

Towards Topographical SLAM§ - *State estimation, simulation, Python*

- 1st place winner of the Informs OR/MS poster competition at the graduate level for my developmental work towards ultralightweight robotic state estimation leveraging rangefinder measurements of local elevation instead of vision.

Information-Optimal Path Planning in Uncertain Environments – *Probability theory, path planning, simulation, SLAM*

- Reimplemented and added collision avoidance considerations to the iCR algorithm from the paper "Active Exploration and Mapping via Iterative Covariance Regulation over Continuous SE(3) Trajectories".

Numerical Optimization Library – *Optimization, MATLAB*

- Built a MATLAB library implementing line search with Wolfe conditions, gradient descent, Newton and Quasi-Newton methods (e.g. BFGS).

Matrix Lie Group-based Navigation Library – *Python, probability theory, git*

- Contributor to Navlie§ and Pymlg§ libraries. Fixed significant oversight in the implementation of the Lie group Iterative Kalman Filter and implemented SL(3) group state matrices for homography-based navigation.

Convolutional Neural Network Image Classifier – *Machine/deep learning, optimization*

- Achieved 90% image classification success rate with original network architecture in a Kaggle competition for classifying low resolution images from the fashion MNIST dataset obfuscated by standard MNIST data.

INTERESTS

Skiing • Rock Climbing • Cycling • Soccer • Simulation development